

Pratikesh Howale

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SUMMARY

Data Scientist and GenAI Developer with hands-on experience in machine learning, deep learning, NLP, and applied LLM systems. Proficient in building end-to-end ML pipelines, RAG-powered applications, and agentic AI systems using LangChain and LangGraph. Skilled in transforming raw data into actionable insights through data preprocessing, feature engineering, and model optimization.

TECHNICAL SKILLS

Languages: Python, SQL

ML / DL Frameworks: Scikit-Learn, TensorFlow, PyTorch, XGBoost

GenAI & LLM: LangChain, RAG, Agentic Chunking

Data & Visualization: NumPy, Pandas, Seaborn, Matplotlib, Tableau, Excel

NLP & AI: NLP, Large Language Models (LLM), Neural Networks, Deep Learning

Tools & Platforms: Streamlit, Jupyter Notebook, Git, Flask

Concepts: Statistics, EDA, Feature Engineering, Hyperparameter Tuning, Model Evaluation

EXPERIENCE

YBI Foundation | October 2024

Data Science Intern — Pune, Maharashtra (Online)

- Completed an intensive 1-month internship focused on Data Science and Machine Learning, applying industry-standard workflows to real datasets.
- Gained hands-on experience with data preprocessing pipelines, model evaluation metrics, and algorithm optimization techniques.
- Delivered 3 end-to-end machine learning projects, applying classification and regression algorithms to practical business problems.

DATA MIND | October 2022 – March 2023

Data Science Trainee — Hyderabad, Telangana

- Completed a rigorous 6-month training program covering Machine Learning and Deep Learning with hands-on project exposure.
- Developed expertise in data preprocessing, feature engineering, model selection, and performance evaluation using Python.
- Applied cutting-edge ML and DL methods to tackle real-world classification and regression problems across structured datasets.

PROJECTS

AI Customer Support Chatbot with RAG

Python, LangChain, RAG, FAISS, OpenAI, Agentic Chunking, Streamlit | June 2026 – July 2026

- Built a RAG-powered support chatbot grounded in a custom knowledge base (FAQ, return policy, product manual) with source citations on every answer.
- Implemented agentic chunking using GPT-3.5-turbo to split knowledge base documents at natural topic boundaries, replacing fixed-size character splitting for higher retrieval quality.
- Integrated ConversationBufferMemory to maintain full multi-turn conversation history, enabling accurate follow-up question handling across sessions.

Credit Score Prediction for Vehicle Loan Risk Assessment

Python, Scikit-Learn, XGBoost, Pandas | Oct 2022 – Mar 2023

- Engineered and preprocessed a large-scale credit dataset with 279,856 records and 15 features using NumPy and Pandas to assess individual creditworthiness.
- Implemented and benchmarked 6 ML algorithms: Linear Regression, Decision Tree, Random Forest, KNN, XGBoost, and SVM; developed an ensemble stacking model.
- Achieved Combined Model Accuracy of 29.35 MSE via ensemble stacking, outperforming standalone models (Random Forest: 41.45 MSE).
- Applied hyperparameter tuning to optimize model performance and reduce prediction error across all algorithms.

Travel Ticket Cancellation Prediction for Risk Management

Python, Scikit-Learn, XGBoost, Pandas | Sep 2023 – Oct 2023

- Built a binary classification system to predict ticket cancellations, enabling proactive risk management and revenue protection strategies.
- Handled class imbalance using one-hot encoding and undersampling techniques; applied StandardScaler for feature normalization.
- Trained and evaluated Decision Tree, Random Forest, KNN, and XGBoost classifiers, achieving over 94% accuracy on the test set.
- Optimized model hyperparameters using GridSearchCV; evaluated performance using precision, recall, F1-score, and confusion matrices.

MetaCritic Movies Rating Analysis Dashboard

Tableau | Sep 2024 – Oct 2024

- Analyzed a dataset of 16,000+ movies spanning 1910–2024 from MetaCritic to uncover trends in ratings, genres, and release patterns.
- Built interactive Tableau dashboards featuring KPIs, average ratings by year, top-rated years, genre popularity trends, and calculated fields.
- Delivered actionable insights on genre performance and audience preferences through advanced filtering and visualization techniques.

EDUCATION

Sinhgad Institute of Management | October 2022

Master of Computer Applications (MCA) — Pune, Maharashtra

Sinhgad College of Science | April 2019

Bachelor of Science – Computer Science (B.Sc. CS) — Pune, Maharashtra

CERTIFICATIONS

- Data Science Internship – 30 Days (YBI Foundation)
- Deep Learning A-Z 2024: Neural Networks & AI
- 2024 Tableau Certified Data Analyst Training
- The Data Analyst Course: Complete Data Analyst Bootcamp
- Data Analysis Essentials Using Excel
- The Ultimate Flask Course